



Zoox offers free rides

Autonomous car-driving company, Zoox, will begin offering free rides to the members of the public in California. <https://digit.in/jan19-9>



Planets of riches

Astronomers have discovered a bunch of planets or 'Super-Earths' that are filled with rubies and sapphires. <https://digit.in/jan19-10>

cache redesign and also implies that we might see a dedicated component for AI workloads. The other improvements that Golden Cove promises include improved Single Threaded performance and networking/5G performance along with additional security features.

Stations have been increased and the L1 Cache has been bumped up by 50%. In Coffee Lake, this was at 32 KB so Sunny Cove will have 48 KB for the same. The L2 cache size is also being bumped up depending on the actual SKU. The Core-i7 and Core-i9 SKUs will see the largest bump in the main-

in with the larger cache sizes because the two go hand in hand. The Address Generation Units (AGU) has also been bumped up from three to four.

General Purpose - Smarter

This is more to do with reorganising the elements on the CPU die to ensure that the different components can talk to each other quicker i.e. reducing latency. Often, communication between different components takes place over multiple clock cycles depending on how far the two components are situated. By moving the components that frequently communicate with each other closer, there can be a minimal performance increment.

Intel also announced improvements to the branch predictor logic, though to what extent, we don't know.

Special Purpose

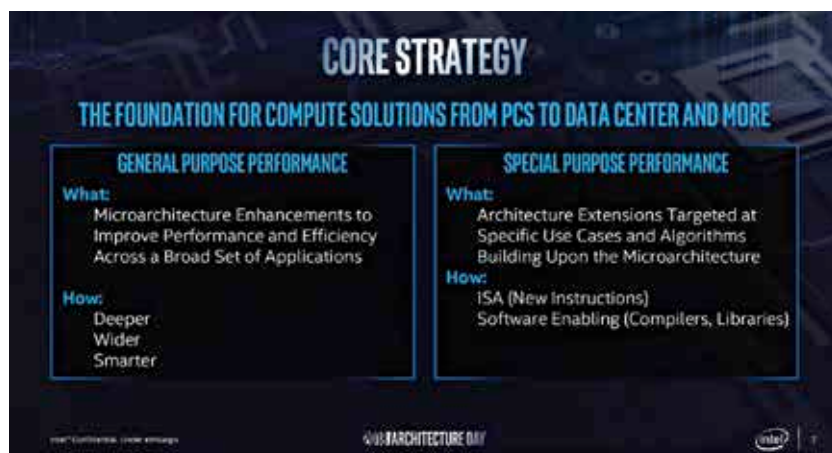
Intel unveiled a few example areas of interest for improving performance. These include

- AI/ML
- Cryptography
- Compression/decompression,
- Communication/networking,
- General SIMD and Special SIMD
- Multi-Threaded & multiple agent processing.

The exact changes might be revealed closer to launch though some of these changes are slated for Golden Cove so we might not see all of them being introduced in Sunny Cove.

THAT'S NOT ALL FOLKS

Intel also unveiled the next generation of IGP i.e. the Gen 11 IGP which when clubbed with Sunny Cove will be called Ice Lake. It features much higher performance owing to new features such as tile-based rendering, and due to the much higher density of cores. Some of which will be seen in Intel's discrete graphics card that is set for a 2020 launch. Lastly, there was the Foveros update, an interposer technology that boasts of significant improvements over the existing EMIB technology. So, sunny days ahead? We'll wait and see how it turns out. 



The approach to improve general and special purpose performance

SUNNY COVE

The entire set of improvements being brought to the table by Sunny Cove are categorised by Intel into two parts. The first being 'General Purpose Performance' and the other being 'Special Purpose Performance'. Intel has a three-pronged strategy to increase General Purpose performance summarised by the words Deeper, Wider and Smarter. The end goal is to increase the IPC (instructions per clock) and it's being done by making the cores 'wider' i.e. allowing more instructions to be executed per clock, 'deeper' i.e. increased parallelism per clock and 'smarter' i.e. improved manner of data delivery to the processor.

As for the Special Purpose performance increase, it would entail incorporating architectural changes and introducing new instruction sets to better leverage the hardware.

General Purpose - Deeper

Intel has increased the size of multiple buffer elements and cache memories. The size of the Reorder Buffer, Load Buffer, Store Buffer and Reservation

stream desktop segment while that for Core-i3 and i5 will be modest.

Increasing the L1 Cache would mean that more data can be stored and as a result, there will be fewer cache misses. A cache miss happens when an instruction looks for a data value in the cache but does not find the same because it has already been purged. Reducing this helps more instructions to be executed per clock cycle. Then there's the micro-op (uOp) cache which has been bumped up from a 2048-entry design. The last element to be buffed is the Translation Lookaside Buffer (TLB) that helps retain a larger number of addresses of data that were recently accessed by the CPU.

General Purpose - Wider

This part is pertinent to increasing the bandwidth of different elements within the processor. First up is the allocation which has been increased from four to five, allowing the Reorder Buffer to push five instructions per clock. The execution ports have also been increased from eight to ten, ensuring that more instructions can exit the scheduler. The increased bandwidth ties